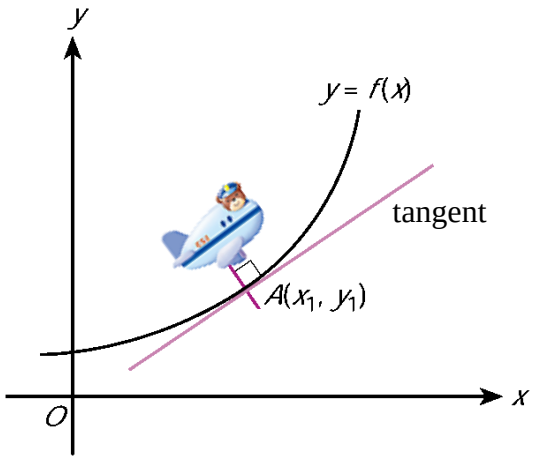
Rules of Differentiation（微分法則）

還記得我們可以用微分去求得切線的斜率 (slope of tangent) 嗎？



$$\text{Slope of tangent} = \frac{dy}{dx}$$

[$\frac{dy}{dx}$ 亦可寫成 $f'(x)$ 或是 y']

但若每次微分都以基本定律 (first principle) 去做的話，那豈不是很麻煩嗎？

e.g. Find the slope of the tangent of $y = x^3$ at $x = 2$.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{(x+\Delta x)^3 - x^3}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{[x^3 + C_1^3 x^2(\Delta x) + C_2^3 x(\Delta x)^2 + (\Delta x)^3] - x^3}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{3x^2(\Delta x) + 3x(\Delta x)^2 + (\Delta x)^3}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \left[3x^2 + 3x(\Delta x) + (\Delta x)^2\right]$$

$$= 3x^2 + 3x(0) + (0)^2 \qquad = 3x^2$$

$\therefore$ At $x = 2$, slope of tangent $= 3(2)^2 \quad = 12$

睇到頭都暈 ~ ~





所以我們一定要緊記以下的法則，那便能夠『秒殺』所有微分！！！

Not for sale

Rules of Differentiation (Power Rule)

Rule No. 1: **Power Rule 冪法則**

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

e.g. ***1.*** $\frac{d}{dx}(x^2) = 2x$

2. $\frac{d}{dx}(x^3) = 3x^2$



3. $\frac{d}{dx}(x^4) =$

4. $\frac{d}{dx}(x^5) =$

5. $\frac{d}{dx}(x^{10}) =$

6. $\frac{d}{dx}(x) =$

7. $\frac{d}{dx}(x^{-1}) =$

8. $\frac{d}{dx}(x^{-2}) =$

9. $\frac{d}{dx}(x^{\frac{1}{2}}) =$

10. $\frac{d}{dx}(x^{-\frac{2}{3}}) =$

Not for sale

Rules of Differentiation (Constant Rule)

Rule No. 2 : **Constant Rule (*k* is a constant 常數)**

$$\frac{d}{dx}(k) = 0$$

e.g. ***1.*** $\frac{d}{dx}(3) = 0$

2. $\frac{d}{dx}(-1) = 0$

Rule No. 3 : **Constant Multiple Rule (*k* is a constant)**

$$\frac{d}{dx}[k \cdot f(x)] = k \cdot \frac{d}{dx}[f(x)]$$

e.g. ***1.*** $\frac{d}{dx}(3x^2) = 3 \cdot \frac{d}{dx}(x^2) = 3 \cdot 2x = 6x$

2. $\frac{d}{dx}(2x^4) = 2\frac{d}{dx}(x^4) = 2 \cdot (4x^3) = 8x^3$



3. $\frac{d}{dx}(5x^7) =$

4. $\frac{d}{dx}(\frac{1}{2}x^{10}) =$

5. $\frac{d}{dx}(2x^{-2}) =$

Not for sale

Rules of Differentiation (Sum and Different Rule)

Rule No. 4 : **Sum and Different Rule** 加及減法則

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$$

e.g. ***1.*** $\frac{d}{dx}(x^2+1) = 2x + 0 = 2x$

2. $\frac{d}{dx}(3x^3 - 2x) = 9x^2 - 2$



3. $\frac{d}{dx}(2x^2 - 3x + 4) =$

4. $\frac{d}{dx}(x^{10} + 9x^8 + 7x) =$

5. $\frac{d}{dx}(x^{-2} - 3x^{-5} + 5x^{-8}) =$

6. $\frac{d}{dx}(x + \sqrt{x}) =$

7. $\frac{d}{dx}(x^2 - 2 + \frac{1}{x^2}) =$

Not for sale



Rules of Differentiation (Product Rule)

Rule No. 5: **Product Rule 積法則**

$$\frac{d}{dx}[f(x)\cdot g(x)] = g(x)\cdot\frac{d}{dx}[f(x)] + f(x)\cdot\frac{d}{dx}[g(x)]$$

e.g. ***1.*** $\frac{d}{dx}(x+1)(x+2) =$ $= (x+2)\cdot\frac{d}{dx}(x+1) + (x+1)\cdot\frac{d}{dx}(x+2)$

$= (x+2)\cdot(1) + (x+1)\cdot(1)$

$= 2x+3$



2. $\frac{d}{dx}(4x+1)(3x-2) =$

---

3. $\frac{d}{dx}(x^2+3)(2x+1) =$

Not for sale

Rules of Differentiation (Quotient Rule)

Rule No. 6: **Quotient Rule** 商法則

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\cdot\frac{d}{dx}[f(x)] - f(x)\cdot\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

阿仔 / 阿媽

秘訣：D 完阿仔 D 阿媽，分母 _____ ！

e.g. ***1.***

$$\frac{d}{dx}\left(\frac{x+1}{x+2}\right) = \frac{(x+2)\cdot\frac{d}{dx}(x+1) - (x+1)\cdot\frac{d}{dx}(x+2)}{(x+2)^2}$$

$$= \frac{(x+2)\cdot(1) - (x+1)\cdot(1)}{(x+2)^2}$$

$$= \frac{1}{(x+2)^2}$$



2. $\frac{d}{dx}\left(\frac{3x+4}{2x+3}\right) =$

3. $\frac{d}{dx}\left(\frac{x^2}{2x+1}\right) =$

Not for sale



Rules of Differentiation (Chain Rule)

Differentiation of Composite Functions (複合函數)

e.g. 若 $g(x)=x^{100}$ 及 $f(x)=2x+3$

則 $g(f(x))=(2x+3)^{100}$ 為複合函數 (composite function) 。

若要微分複合函數，我們需要運用以下法則：

Rule No. 7: **Chain Rule 鏈式法則**

$$\frac{d[g(f(x))]}{dx}=\frac{d[g(f(x))]}{d[f(x)]}\cdot\frac{d[f(x)]}{dx}$$

或表示成 $\dfrac{dy}{dx}=\dfrac{dy}{du}\cdot\dfrac{du}{dx}$

e.g. Differentiate $y=(2x+3)^{100}$ with respect to x.

Wrong approach

$$\frac{dy}{dx}=\frac{d}{dx}(2x+3)^{100}$$

$$=\frac{d}{dx}(2^{100}x^{100}+C_1^{100}2^{99}x^{99}\cdot 3+\ldots)$$



Correct approach

$$\frac{dy}{dx}=\frac{d(2x+3)^{100}}{dx}$$

$$=\frac{d(2x+3)^{100}}{d(2x+3)}\cdot\frac{d(2x+3)}{dx}$$

$$=100(2x+3)^{99}\cdot(2)$$

$$=200(2x+3)^{99}$$



秘訣：D 完指數 D 括號！

Note: 普遍來說， $\dfrac{d(ax+b)^n}{dx}=\dfrac{d(ax+b)^n}{d(ax+b)}\cdot\dfrac{d(ax+b)}{dx}=a\cdot n(ax+b)^{n-1}$

Not for sale

Rules of Differentiation (Chain Rule)



More examples:

1. $\frac{d}{dx}(2x+3)^{200} =$

2. $\frac{d}{dx}(x^2+2x+3)^{100} =$

3. $\frac{d}{dx}\sqrt{2x+3} =$

4. $\frac{d}{dx}\sqrt{(x+2)(2x-3)} =$

5. $\frac{d}{dx}\sqrt{\frac{2x-3}{3x+4}} =$

Not for sale

Rules of Differentiation (Inverse Function Rule)

Rule No. 8: **Inverse Function Rule** 反函數法則

$$\frac{dy}{dx} = 1 \div \frac{dx}{dy}$$

e.g. If $x = y^3 - 1$, find $\frac{dy}{dx}$.

Normal approach

$$x = y^3 - 1$$

$$y = (x+1)^{\frac{1}{3}}$$

$$\frac{dy}{dx} = \frac{d}{dx}(x+1)^{\frac{1}{3}}$$

$$= \frac{1}{3}(x+1)^{-\frac{2}{3}} \cdot (1) \quad = \frac{1}{3(x+1)^{\frac{2}{3}}}$$

Smart approach

$$\frac{dx}{dy} = \frac{d}{dy}(y^3 - 1)$$

$$= 3y^2$$

$$\frac{dy}{dx} = \frac{1}{3y^2}$$

$$= \frac{1}{3(x+1)^{\frac{2}{3}}}$$

Note: 其實我們可以任何主項去微分函數，例如：

若 $\begin{cases} x = 2t \\ y = 1 - t^2 \end{cases}$ (參數式 parametric form), 則

$$\frac{dx}{dt} = \frac{d}{dt}(2t) = 2 \qquad \text{及} \qquad \frac{dy}{dt} = \frac{d}{dt}(1-t^2) = -2t$$

所以 $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ (Chain Rule 鏈式法則)

$= \frac{dy}{dt} \cdot (1 \div \frac{dx}{dt})$ (Inverse Function Rule 反函數法則)

$= -2t \cdot (\frac{1}{2}) \quad = -t$

Not for sale

Rules of Differentiation (Inverse Function Rule)



More examples:

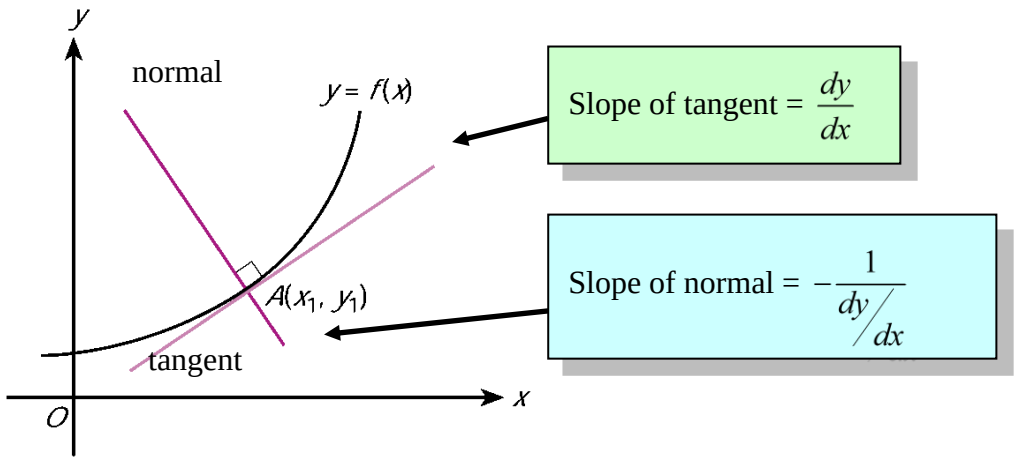
1. If $x = \dfrac{2y+1}{y}$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of y.

2. If $x = \dfrac{y}{\sqrt{2y-1}}$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of y.

Not for sale

Equation of Tangents and Normals

我們可以用微分去求得切線 (tangent) 及法線 (normal) 的斜率：



加上在已知斜率 ***m*** 及一個座標 **(x_1, y_1)** 的情況下，我們就可以求得直線方程 (亦即是切線或法線的方程)：

$$y - y_1 = m(x - x_1)$$

e.g. Find the equation of the tangent of $y = x^2 + 1$ at (2, 5).

$$y = x^2 + 1$$

$$\frac{dy}{dx} = 2x$$

When $x = 2$, $\frac{dy}{dx} = 2(2) = 4$

Step 1:
首先求得切線 (tangent) 或法線 (normal) 的斜率

The equation of the tangent is

$$y - 5 = 4(x - 2)$$

$$y - 5 = 4x - 8$$

$$4x - y - 3 = 0$$

Step 2:
用 $y - y_1 = m(x - x_1)$ 去求切線或法線的方程

Not for sale

Equation of Tangents and Normals

[A.Math 1991 Paper I Q.6 Modified]

Let C be the curve $y = x + \frac{1}{x}$, where $x \neq 0$. $P(1, 2)$ lies on C.

Find equations of the tangent and normal to C at P. ***(7 marks)***

Not for sale

HKCEE Questions (level 1)

1. ***[A.Math 1985 Paper I Q.1]***

Let f(x) = $x\sqrt{1-x^2}$. Find the value of f'($\frac{1}{2}$). ***(5 marks)***

2. ***[A.Math 1997 Paper I Q.1]***

Let f(x) = $\sqrt{3+x^2}$, find f'(–1). ***(3 marks)***

Not for sale

HKCEE Questions (level 1)

3. ***[A.Math 2001 Q.1]***

Find $\frac{d}{dx}\left(\frac{x^2}{2x+1}\right)$. ***(3 marks)***

Not for sale

HKCEE Questions (Level 1)

4. ***[A.Math 2008 Q.6]***

Find the equation of the tangent to the curve $y = \dfrac{3x}{x^2 + 2}$ at the point (2, 1).

(5 marks)

Not for sale

HKCEE Questions (Level 1)

5. ***[A.Math 2002 Q.2]***

Find the equation of the tangent to the curve C: $y = (x-1)^4 + 4$ which is parallel to the line $y = 4x + 8$. ***(4 marks)***

Not for sale

HKCEE Questions (Level 2)

6. ***[A.Math 1996 Paper I Q.6]***

Find the equations of the two tangents to the curve C: $y = \frac{6}{x+1}$ which are parallel to the line $x + 6y + 10 = 0$. ***(7marks)***

Not for sale

HKCEE Questions (Level 2)

7. ***[A.Math 1982 Paper I Q.5]***

Show that the tangents to the curve $y = x^3 - 9x^2 + 30x + 4$ cannot be parallel to the x-axis. ***(6 marks)***

Not for sale

HKCEE Questions (level 2)

8. ***[A.Math 1989 Paper I Q.3]***

Find the coordinates of the two points on the curve $y = x^3$ at which the tangents to the curve have a slope of $\frac{3}{4}$.

Hence find the equations of the two tangents to the curve $y = x^3$ which are parallel to the line $3x - 4y = 0$. ***(5 marks)***

Not for sale

HKDSE Questions (level 2)

9. ***[M1 Practice Paper Q.4]***

Consider the curve C: $y = x(2x-1)^{\frac{1}{2}}$, where $x > \frac{1}{2}$.

(a) Find $\frac{dy}{dx}$.

(b) Using (a), find the equations of the two tangents to the curve C which are parallel to the straight line $2x - y = 0$. ***(6 marks)***

Not for sale

HKDSE Questions (level 2)

10. ***[M1 2015 Q.7]***

Consider the curve C : $y = x\sqrt{2x^2 + 1}$.

(a) Find $\dfrac{dy}{dx}$.

(b) Two of the tangents to C are perpendicular to the line $3x + 17y = 0$.

Find the equations of the two tangents. ***(7 marks)***

Not for sale

Not for sale

HKCEE Questions (Level 2)

13. ***[A.Math 2004 Q.9]***

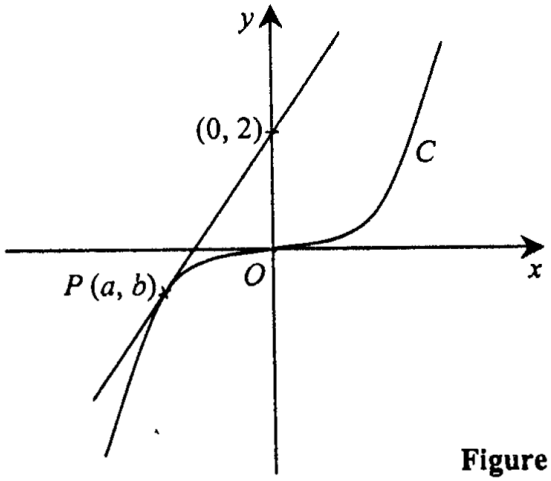


Figure 3

In Figure 3, $P(a, b)$ is a point on the curve C: $y = x^3$. The tangent to C at P passes through the point (0, 2).

(a) Show that $b = 3a^3 + 2$.

(b) Find the values of a and b.

(6 marks)

Not for sale